

PROJECT KRATOS - ENC Inductions

Week 3, Q1

Given: $L_1 = 2m$, $L_2 = 1.5m$.
Base fixed at $(0,0)$, xy-plane.

• Max Reach of Robot's Arm:-

$$|L_1 + L_2| = 2.0m + 1.5m = 3.5m$$

• Min. Reach of Robot's Arm:-

$$|L_1 - L_2| = 2.0 - 1.5 = 0.5m$$

We know,

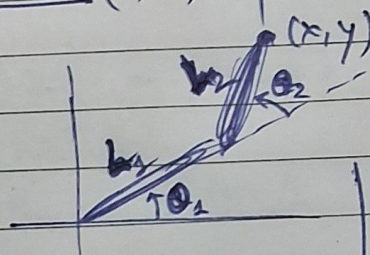
INVERSE KINEMATICS

(IK)

- Cartesian coord (x, y, z) input
- req^d joint parameters (θ, ℓ) output.

FORWARD KINEMATICS (FK)

- Angles, lengths as input
- Cartesian coord (x, y, z) output



$$\cos \theta_2 = \frac{x^2 + y^2 - L_1^2 - L_2^2}{2L_1L_2} = \frac{x^2 + y^2 - 6.25}{6}$$

$$\theta_2 = \cos^{-1} \left(\frac{x^2 + y^2 - 6.25}{6} \right)$$

$$\theta_1 = \tan^{-1} \left(\frac{y}{x} \right) - \tan^{-1} \left(\frac{L_2 \sin \theta_2}{L_1 + L_2 \cos \theta_2} \right)$$

$$x_{calc} = L_1 \cos \theta_1 + L_2 \cos(\theta_1 + \theta_2)$$

$$y_{calc} = L_1 \sin \theta_1 + L_2 \sin(\theta_1 + \theta_2)$$

TARGET 1:-

$(2m, 1m)$

$$d = \sqrt{2^2 + 1^2} = \sqrt{5}$$

Since $|L_1 - L_2| < d < |L_1 + L_2|$
 $0.5m < 2.236m < 3.5m$

Reachable.

(IK) :- $\cos \theta_2 = \frac{(5) - 6.25}{6} = -0.20833$; $\theta_2 = 1.7806 \text{ rad}$

ELBOW UP CASE :- $\theta_1 = \tan^{-1} \left(\frac{1}{2} \right) - \tan^{-1} \left(\frac{1.5 \sin(1.7806)}{2 + 1.5 \cos(1.7806)} \right)$; $\theta_1 = -0.2520 \text{ rad}$

ELBOW DOWN CASE $\Rightarrow \theta_2 = -1.7806 \text{ rad}$ & $\theta_1 = 0.4636 + 0.7156 = 1.1792 \text{ rad}$

(FK) :- $x = 2 \cos(-0.2520) + 1.5 \cos(-0.2520 + 1.7806) = 2m$ ✓
 $y = 2 \sin(-0.2520) + 1.5 \sin(-0.2520 + 1.7806) = 1m$ ✓

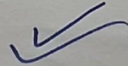
[Verification]

TARGET 2:- $d = \sqrt{3^2 + 0^2} = 3\text{m}$. Since $0.5\text{m} < 3\text{m} < 3.5\text{m}$, Reachable
(3m, 0m)

(FK) $\cos \theta_2 = \frac{9 - 6 \cdot 25}{6} = 0.4833$; $\theta_2 = 1.1034\text{rad}$
ELBOW UP CASE $\theta_1 = \tan^{-1}\left(\frac{0}{3}\right) - \tan^{-1}\left(\frac{1.5 \sin(1.1034)}{2 + 1.5 \cos(1.1034)}\right) = 0.0 - 0.4605 = -0.4605\text{rad}$

ELBOW DOWN CASE $\theta_2 = -1.1034\text{rad}$ & $\theta_1 = 0.0 + 0.4605$, $\theta_1 = 0.4605\text{rad}$

(FK) $x = 2 \cos(-0.4605) + 1.5 \cos(-0.4605 + 1.1034) = 3\text{m} \checkmark$
 (Verification) $y = 2 \sin(-0.4605) + 1.5 \sin(-0.4605 + 1.1034) = 0\text{m} \checkmark$



TARGET 3:- $d = \sqrt{0^2 + (3.4)^2} = 3.4\text{m}$. Since $0.5\text{m} < 3.4\text{m} < 3.5\text{m}$, Reachable
(0m, 3.4m)

(FK) $\cos \theta_2 = \frac{(3.4)^2 - 6 \cdot 25}{6} = 0.8830$; $\theta_2 = 0.4843\text{rad}$
ELBOW UP $\theta_1 = \tan^{-1}\left(\frac{3.4}{0}\right) - \tan^{-1}\left(\frac{1.5 \sin(0.4843)}{2 + 1.5 \cos(0.4843)}\right)$
 $= 1.5708 - 0.2069$; $\theta_1 = 1.3639\text{rad}$

ELBOW DOWN $\theta_2 = -0.4843\text{rad}$ & $\theta_1 = 1.5708 + 0.2069$, $\theta_1 = 1.7777\text{rad}$

(FK) $x = 2 \cos(1.3639) + 1.5 \cos(1.3639 + 0.4843) = 0\text{m} \checkmark$
 (Verification) $y = 2 \sin(1.3639) + 1.5 \sin(1.3639 + 0.4843) = 3.4\text{m} \checkmark$

